Concrete-Complexity Cryptanalysis of the 2026 Poseidon1 / KoalaBear Challenge Suite

Ömer Demirel*

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Abstract

We report a verifier-exact reimplementation and a concrete-complexity analysis of the four 2026 challenge predicates of the Poseidon Cryptanalysis Initiative, instantiated over the KoalaBear field $p = 2^{31} - 2^{24} + 1$ with the Poseidon1 permutation ($\alpha = 3$, width $t = 16$). Our contributions are: (i) an independent, golden-vector-validated Python/Rust implementation for the default vendored Cauchy/Toeplitz+Grain instance, with explicit MDS scope (the organizers accept arbitrary admissible MDS matrices); (ii) a reproduction and verifier-instance calibration of the CICO-2 resultant degree law $\deg R = 3^{2R_F+R_P}$ from the resultant-attack literature, showing that skip-first-rounds is a constant-factor refinement rather than a prerequisite on this instance; (iii) a root-first resultant model for the zero-test predicate, showing that the fixed-point feedforward system has the same CICO-like degree order and giving exact verifier-accepted reduced-round witnesses; (iv) an empirical refutation of an exploitable low-order multiplicative-character bias in the density predicate, with a corrected success-rate baseline of 2^{-48}; and (v) a no-go cost table for the full-strength ($R_F = 8, R_P = 20$) partial-collision ladder. We do *not* claim a solve of any full-round ($R_F = 6$) instance; we map the measured cost of reaching one and *port and measure* the one structural advance often suggested for it — an amortized resultant — finding it does *not* bring $R_F = 6$ within practical reach: priced end-to-end, the cost is dominated by the build-once bivariate construction (time $\sim \delta^{3.6}$, memory $\approx 4\delta^2$), not the per-node resultant ($< 0.1\%$ of the total). Under this model the end-to-end build is on the order of 10^3 core-years and ≈ 1.0 TiB at $R_F = 6/R_P = 6$, rising to ≈ 9.2 TiB at $R_F = 6/R_P = 7$ and, beyond the current $R_F = 6/R_P = 8$ public record, to $\approx 10^8$ core-years and ≈ 745 TiB (out-of-core in every case) at $R_F = 6/R_P = 9$. The port is exact at reduced rounds but its full-round economics are a measured obstruction. Every new measured claim is reproducible, and projections are explicitly labelled (Section 8).

1 Introduction

The Poseidon Cryptanalysis Initiative poses four families of challenge predicates over an arithmetization-oriented hash. Unlike bit-oriented primitives, Poseidon's power-map S-boxes and partial rounds turn each challenge into an explicit polynomial system over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$, so the relevant attack tooling is algebraic: resultants, GCDs, and structured root-finding. This paper does not claim a break of a full-round instance. It reports the *measured* complexity landscape of the 2026 suite, several corrections to widely-held assumptions about the instance, and one new structural reduction for the zero-test predicate.

*Snowfall Finance. Email: `omer@progfi.xyz`. An independent concrete-complexity study of the 2026 Poseidon Cryptanalysis Initiative challenge suite. Source code and artifacts are released publicly under the MIT license; see Section 8.

Relation to prior work. The Initiative's survey of attacks on Poseidon and Poseidon2 [5] is the reference baseline for algebraic attacks on these instances. Its algebraic-attack section is organized around interpolation and Gröbner-basis attacks, with CICO interpolation degree d^R and Gröbner lex-basis estimates such as $C_u = d^{\chi R_F + R_P - 2\chi + 1}$ (Faugère–Perret / BSGL20) and $D = d^{\chi R_P + R_F}$ (Steiner). It does not give a resultant baseline for the 2026 KoalaBear CICO-2, density, or zero-test predicates. The closest prior resultant work is Bak, Bariant, Bœuf, Hostettler, and Jazeron [4], who claim the Ethereum Foundation's earlier bounties on *Poseidon2* instances with resultant-based algebraic attacks, including an amortized bivariate resultant that shares the elimination across all interpolation nodes. Our target differs: the 2026 Poseidon Cryptanalysis Initiative's *Poseidon1* suite with four predicates (partial collision, CICO-2, density, zero-test). The CICO degree law used below is therefore not presented as a new asymptotic theorem; our contribution is its exact reproduction on the default KoalaBear/Poseidon1 verifier instance and the resulting concrete calibration. Recent Gröbner-basis work by Grassi, Koschatko, and Rechberger [6] studies related Poseidon/Neptune algebraic attacks through subspace trails and mode-of-operation effects; our zero-test route is instead a resultant-based fixed-point model. Relative to the survey, the new contribution here is this executable resultant treatment of the 2026 Poseidon1/KoalaBear predicates — especially the root-first zero-test model — plus measured reduced-round witnesses and full-round obstruction data, not a full-round solve. We also port the amortized resultant idea of [4] to this zero-test route and validate it at reduced rounds; the full-round consequence is still a measured projection, not a bounty solve (Section 10).

Scope and honesty. Every new measured quantitative claim is reproducible from the commands and artifacts listed in Section 8. We separate results from projections explicitly: Table 2 marks each contribution as *measured*, *projection*, or *organizer-confirmed scope*, and we claim no full-round solve. The exposition is self-contained: challenge predicates, reductions, and cost models are defined before use; the repository supplies reproducibility evidence rather than replacing the arguments.

Security interpretation. The survey defines correlation intractability as the infeasibility of finding inputs M_j and outputs V_j with $h(M_j) = V_j$ and a target relation $R(M_1, \ldots, M_r, V_1, \ldots, V_r) =$ true faster than $|\mathbb{F}_p|^r$ ([5], Definition 3). The zero-test predicate instantiates such a relation: M encodes a low-degree polynomial and the hash output V (viewed in $\mathbb{F}_{p^2}$) is required to be a root of that polynomial. This output-is-a-root shape is the kind of algebraic relation that appears in polynomial-commitment / FRI-style proof-of-knowledge soundness checks, which is why a concrete algebraic model of it is of independent interest. Our root-first resultant model produces exact reduced-round witnesses for this relation together with a concrete obstruction for the full-round target.

2 Preliminaries

Field. KoalaBear $p = 2^{31} - 2^{24} + 1 = 2130706433$, with $p - 1 = 2^{24} \cdot 127$; the large power of two makes $\mathbb{F}_p$ NTT-friendly. The S-box is $x \mapsto x^3$.

Permutation. Poseidon1 with width $t = 16$: an optional initial linear layer (for CICO), then $R_F/2$ full rounds, R_P partial rounds (cubing lane 0 only), then $R_F/2$ full rounds. The default vendored verifier instance studied here uses the Cauchy/Toeplitz matrix $M_{ij} = (i - t - j)^{-1} \bmod p$ and round constants from the Grain LFSR of the Poseidon specification. Organizer clarification on 2026-06-17 confirmed that submissions may instead use arbitrary MDS matrices satisfying the invariant-subspace requirements, and that showing some admissible matrices are more vulnerable may qualify for a bonus. Unless explicitly stated otherwise, the executable witnesses and concrete-complexity measurements in this paper target the default Cauchy/Toeplitz `poseidon-tools` instance.

The four predicates. *(a) Partial collision:* full-strength $R_F = 8, R_P = 20$, compression mode; collide the first $q \in \{3, \dots, 7\}$ outputs of $H(\text{seed}, X)$. *(b) CICO:* $R_F = 6$, $R_P \in \{6, 8, 10\}$, `permutation_plus_linear`; two-in/two-out ($k = 2$). *(c) Density:* $R_F = 6, R_P > 5$, $d = 2$ attacker-chosen zero positions, decode each output through $x^{(p-1)/16}$. *(d) Zero-test:* $R_F = 6, R_P > 5$, a degree-≤ 7 polynomial P over $\mathbb{F}_{p^2} = \mathbb{F}_p[x]/(x^2 - 3)$ whose first hashed output is a root of P. The Initiative page accessed on 2026-06-17 lists CICO $R_F = 6/R_P = 6$ and $R_F = 6/R_P = 8$ as verified, leaves CICO $R_P = 10$ listed, and reports the current zero-test record as $R_F = 6/R_P = 8$ [8]. The survey's 31-bit 128-bit-security row refers to Poseidon2 over the Mersenne field M31; the 2026 Initiative separately defines the Poseidon-31 Poseidon1/KoalaBear instance studied here, and that instance is the one used for all executable claims below.

3 Verifier-exact reimplementation

We reimplemented the primitive and all four acceptance predicates independently in Python (reference) and Rust (Montgomery-arithmetic core), and cross-checked them against published golden vectors.

Observation 1 (Instance / MDS scope)**.** *The challenge admits any MDS matrix satisfying the no-invariant-subspace-trail conditions; the Plonky3 width-16 circulant is an admissible example, and the vendored* `poseidon-tools` *repository ships the Cauchy/Toeplitz matrix $M_{ij} = (i - t - j)^{-1}$ with Grain constants as its default test vector. Because acceptance is instance-specific, a solution computed for one admissible MDS does not by itself satisfy a verifier pinned to another, so a submission must fix and publish its exact MDS and verify against that instance. The two matrices differ in a documented way: the circulant* fails *the Poseidon minimal-polynomial criterion (*$16 \mid p - 1$*) yet* passes *the reference implementation's* `verify_mds_matrix` *checks, so both are accepted by the admissibility gate. Our golden suite anchors on (i) the published Plonky3 permutation vector (round-schedule check), (ii) the default Cauchy+Grain value* $\text{perm}(0^{16})_0 = 1393439926$*, and (iii) the published $t = 3$ partial collision (accept $t \leq 3$, reject $t = 4$). All executable witnesses and concrete-complexity measurements below use the default Cauchy/Toeplitz instance; the search for an admissible but more attackable MDS is a separate open direction.*

4 CICO-2: reproduced resultant degree law

We model CICO-2 as a bivariate system in two free input coordinates X, Y and eliminate Y by a resultant $R(X) = \text{Res}_Y(\text{out}_0, \text{out}_1)$, whose $\mathbb{F}_p$-roots back-substitute to solutions.

Claim 1 (Verifier-instance degree reproduction)**.** *For the as-built bivariate system (no skip-first-rounds),* $\deg R = 3^{\,2R_F + R_P}$.

This was measured exactly at $(R_F, R_P) \in \{(2, 0), (2, 1), (2, 2)\}$, giving $\deg R = 81, 243, 729$. The naive Bézout product would give $3^{2(R_F + R_P)}$; the measured law is a constant factor $3^2 = 9$ smaller, equal to the skip-first-rounds ideal degree $D_I = 3^{2(R_F - 1) + R_P}$ times 9. Consequently skip-first-rounds buys a *constant* $9\times$, not an R_P-dependent speedup: it is a refinement, not a prerequisite. This matches the resultant-based CICO analysis of Bak et al. [4]; the contribution here is the default-verifier reproduction, exact KoalaBear calibration, and downstream cost model. For the challenge instances this yields the as-built root-find degrees 3^{12+R_P}, i.e. $2^{28.5}, 2^{31.7}, 2^{34.9}$ for $R_P = 6, 8, 10$.

Brute force is a calibration baseline only. A generic CICO-2 search costs $p^2 = 2^{62}$ permutation evaluations; at the measured native throughput ($\sim 5 \times 10^5$ evaluations/s single-thread) this is $\approx 2.85 \times 10^5$ core-years, confirming that an algebraic attack is mandatory and that a single-machine solve is out of scope.

5 Zero-test: a root-first resultant model

Write a candidate as $P(z) = (z - r)\,G(z)$ with $r \in \mathbb{F}_{p^2}$ and a degree-≤ 6 cofactor G; then $P(r) = 0$ by construction, and the only coupling is the fixed-point condition $r = a_0 := H(\hat{P})_0 \in \mathbb{F}_{p^2}$, where $\hat{P}$ is the flat coefficient vector hashed in compression mode. Crucially, in compression mode the degree-0 coefficient of P enters both $\hat{P}$ and, via the feedforward $\mathrm{out}_i + \mathrm{in}_i$, the output a_0.

Claim 2 (Root-first CICO-like eliminant)**.** *Solving the fixed-point $a_0 = r$ via resultant elimination yields an eliminant of degree $3^{\,2R_F+R_P}$ (CICO-like),* not *the generic two-variable Bézout product $3^{\,2(R_F+R_P)}$.*

This was measured directly (Table 1). The distinguishing rows $R_P \geq 1$ are decisive: at $(R_F, R_P) = (2, 1)$ the eliminant degree is $243 = 3^5$, not $729 = 3^6$; at $(2, 2)$ it is $729 = 3^6$, not $6561 = 3^8$. Thus the zero-test exact solve is of the *same degree order* as CICO, with two extra degrees of freedom in the parameterization. Once the root-first fixed-cofactor reduction is made, the degree value is expected from the CICO-like two-equation/two-variable shape; the novel piece is the root-first resultant formulation and its exact reduced-round verification. This is, to our knowledge, the first resultant-based model of this zero-test predicate. It is complementary to Gröbner-basis treatments of related Poseidon algebraic systems, including subspace-trail attacks [6], which do not give this fixed-point zero-test route.

Table 1: Measured zero-test root-first eliminant degree vs. the two hypotheses.

R_F	R_P	measured	$3^{2R_F+R_P}$ (CICO-like)	$3^{2(R_F+R_P)}$ (generic)
2	1	243	**243**	729
2	2	729	**729**	6561
4	0	6561	**6561**	6561

Exact reduced-round witnesses. We obtained verifier-accepted exact witnesses (residual $(0, 0)$ in $\mathbb{F}_{p^2}$) at $R_F = 4, R_P = 5$ via a fixed-root/variable-cofactor coordinate slice—a degree $3^{13} =$ 1,594,323 eliminant interpolated over 1,594,324 clean samples with one X-root and one Y-candidate, independently re-verified in Python—at $R_F = 4, R_P = 4$ via both coordinate and affine-plane slices, and at $R_F = 2, R_P = 4$ via reproducible split-shard round trips. The generic single-witness success rate of the predicate is $7/p^2 \approx 2^{-59.2}$, so these are genuine algebraic solves, not search hits.

6 Density: no exploitable low-order character bias

The density predicate requires all 16 decoded outputs — decoded through the order-16 multiplicative character $x \mapsto x^{(p-1)/16}$ — to fall in two attacker-chosen residue classes.

Claim 3 (Corrected baseline and measured uniformity)**.** *The brute-force success rate is* $(d/r)^{16} = (2/16)^{16} = 2^{-48}$*, not* 2^{-64}*: each output may hit* either *of the* $d = 2$ *chosen classes. Empirically, the decode-class distribution at* $R_F = 6, R_P = 6$ *is statistically uniform (*$\chi^2 \approx 14$ *on 15 d.o.f.), with no first- or second-order coset bias, and the zero* value *maps to index* 0 *only with probability* $1/p$ *— so no zero position is special.*

This is a (modest) negative result: the natural multiplicative-character shortcut does not exist at the round counts tested, so density requires a genuinely new construction.

7 Partial collision: a full-strength no-go table

The collision instance is full-strength $R_F = 8, R_P = 20$ (a frequent misreading takes it as $R_F = 6$). The birthday cost for $q = 4$ is $\sqrt{p^4} = 2^{62}$. The Cauchy MDS used by the default reference instance is chosen to defeat invariant/subspace trails, and known subspace-trail bounds do not survive the full-round wrapper. We therefore record a no-go cost table rather than an attack: the permutation degree is $3^{28} \approx 2^{44.4}$ and a CICO-style algebraic collision system has ideal degree $3^{34} \approx 2^{53.9}$, both dominated, with no structural lever observed.

8 Reproducibility

All claims reproduce from the repository. The Python guards are `tests/test_golden.py` (18/18), `test_poly.py`, `test_cico_attack.py`, `test_skip_and_gcd.py` (5/5), and `profile_density_decode.py`. The Rust guards in `core_rs` include `golden` (14/14), `poly_golden`, `cico_golden`, `zerotest_golden` (11/11), `zerotest_calibrate`, `zt_amortized_spike`, and `zt_resultant_fast`. The independent verifiers are built against the default Cauchy+Grain instance and agree with the published golden anchors. The full source, the durable JSON artifacts, and the reproduction commands above are released in the accompanying public repository.

9 Summary: results versus projections

Table 2 separates what we measured from what we project, so the status of every contribution is unambiguous to a reviewer.

10 Limitations and open problems

- **The $R_F = 6$ gate (measured: cluster-scale, and *not* closed by amortization).** *Unamortized route.* Exact witnesses here are reduced-round ($R_F \in \{2, 4\}$); we resolve the $R_F = 6$ economics by *directly* measuring the real interpolated-resultant route rather than a proxy. At each of the D interpolation nodes the attack builds the residual pair f_0, f_1 (a symbolic permutation over $\mathbb{F}_p[X]$ in the root-first state) and computes one $\mathrm{Res}_Y(f_0, f_1)$. We measured the per-node cost on the feasible ladder $R_P = 2, \ldots, 5$ (residual degree $3^8, \ldots, 3^{11}$) and confirmed it against a directly-measured $R_F = 6, R_P = 6$ node, where the per-node

Table 2: Status of each contribution.

Item	Status
Cauchy/Grain default-verifier exactness (golden 14/14)	measured
CICO-2 degree law $3^{2R_F+R_P}$ on the default instance	reproduced / measured
Zero-test root-first resultant model	measured
$R_F = 4, R_P = 5$ zero-test witness (residual $(0,0)$)	measured / verified
$R_F = 6$ un-amortized economics ($R_P = 6, \dots, 9$; $R_P = 9$ first record-improving)	measured projection
Amortized-resultant port (exact) [4]	measured / verified
Fast resultant sub-quadratic (exponent ≈ 1.22)	measured
Amortized $R_F = 6$ end-to-end cost ($R_P = 6, \dots, 9$; build-dominated)	measured projection
Custom-MDS acceptance	organizer confirmed; unmeasured attack surface

cost at residual degree 3^{12} is $3.74 + 112.9 = 116.7\,\mathrm{s}$ (build plus resultant). The resultant scales as degree$^{1.96\text{–}2.01}$ because its quotient sequence is generically length-2 ($\overline{q} \approx 2.03$), so the elimination is $O(n^2)$ Euclidean and the Half-GCD (next bullet) does *not* engage on the un-amortized route. Eliminant degree $D = 3^{2R_F+R_P}$ gives: $R_P = 6$ (at or below the public record) $D = 3^{18} \Rightarrow\approx 1.4 \times 10^3$ core-years; $R_P = 7$ (below the current public record) $D = 3^{19} \Rightarrow\approx 3.7 \times 10^4$ core-years. The Initiative page accessed on 2026-06-17 reports the current zero-test record as $R_F = 6/R_P = 8$, so $R_P = 9$ is the first public-record-improving target; the same measured fit projects the un-amortized route to $\approx 9.6 \times 10^5$ core-years at $R_P = 8$ and $\approx 2.4 \times 10^7$ core-years at $R_P = 9$. The line-GCD route is strictly worse (codimension-2: $\sim 1/p$ solutions per line, hence $\sim p \approx 2.1 \times 10^9$ lines at comparable per-line cost). This corrects an earlier proxy that used a Half-GCD stand-in for the elimination and undercounted the per-node cost by $\approx 6\times$; even integrating a Half-GCD as a per-node *constant* ($\sim 7\times$ at this degree) on the un-amortized route leaves the floor far above feasibility already below the current record, and astronomically so for $R_P \geq 9$.

Amortized route. We therefore *ported* the structured/amortized resultant of [4] — which shares the elimination across all D nodes by evaluation–interpolation — to the Poseidon1/KoalaBear zero-test route and validated it exactly: the amortized eliminant matches the per-node oracle coefficient-for-coefficient at $(R_F, R_P) \in \{(2,1),(2,2),(2,4)\}$ and recovers verifier-accepted witnesses. On *real* residuals the fast resultant is sub-quadratic with a *measured* exponent ≈ 1.22 (exact-equal to schoolbook through degree 3^{12}), so the per-node speedup is a growing ratio, not a fixed constant; the resulting per-node cost model reproduces the [4] $R_F = 6, R_P = 4$ Poseidon2 datapoint (312 core-hours) to within 0.5%. Crucially, however, this per-node fast resultant is *not* the dominant cost. Instrumenting all five end-to-end stages on exact-gated reduced rounds (build, multipoint evaluation, resultant, interpolation, root recovery) shows the per-node resultant is $< 0.1\%$ of the total, while the *build-once* bivariate construction scales as $\approx \delta^{3.6}$ (measured, $R^2 = 0.998$) and dominates (64–76%). The end-to-end projection (optimistic skipped degree $\delta = 3^{R_F-1+R_P}$) is therefore $\approx 1.0 \times 10^3$ core-years at $R_F = 6/R_P = 6$ and $\approx 4.1 \times 10^4$ core-years at $R_F = 6/R_P = 7$; at the current-record $R_P = 8$ it is $\approx 2.1 \times 10^6$ core-years, and at the first public-record-improving $R_P = 9$ it is $\approx 1.1 \times 10^8$ core-years. The build stage *alone* is $\approx 6 \times 10^2$ core-years even at $R_P = 6$ with a free eval. The skip does *not* transfer to the root-first route (its kernel directions lie outside the image of the rigid $(z-r)G(z)$ map), so the implemented naive-degree cost is a further $\approx 56\times$. The build-once bivariate is dense, $\approx 4\delta^2$ bytes ($\approx 1.0\,\mathrm{TiB}$ at $R_P = 6$, $\approx 9.2\,\mathrm{TiB}$ at

$R_P = 7$, ≈ 83 TiB at $R_P = 8$, and ≈ 745 TiB at $R_P = 9$), out-of-core in every case. This memory-access wall is independently corroborated by the Ethereum-Foundation-co-authored NTT lower bound of Zhao–Sanso–Vitto–Ding [7], proving the NTT memory-access cost grows at least as $n^{4/3}$. The amortized resultant is thus exact but does *not* bring $R_F = 6$ within reach; the open question is whether an out-of-core / blockwise build can achieve *both* sub-δ^2 memory and sub-$\delta^{3.6}$ time.

$R_F = 6$ **zero-test economics under the default-MDS model.** RP=8 is the current public record on the 2026-06-17 Initiative page; RP=9 is therefore the first public-record-improving target. Figures are projected compute in core-years at the measured per-node and build scalings.

R_P	role	un-amortized (core-yr)	amortized end-to-end (core-yr)	build memory
6	at/below record	$\approx 1.4 \times 10^3$	$\approx 1.0 \times 10^3$	≈ 1.0 TiB
7	below record	$\approx 3.7 \times 10^4$	$\approx 4.1 \times 10^4$	≈ 9.2 TiB
8	current record	$\approx 9.6 \times 10^5$	$\approx 2.1 \times 10^6$	≈ 83 TiB
9	first improving	$\approx 2.4 \times 10^7$	$\approx 1.1 \times 10^8$	≈ 745 TiB

- **Per-node resultant wall (partially addressed).** The dominant cost at scale is the resultant/GCD of degree-$3^{R_F+R_P}$ univariates. We implemented and verified a recursive Half-GCD (validated against the schoolbook GCD on 241/241 random, planted-factor, and degenerate cases). Measured against the Euclidean baseline it crosses over near degree 6.5×10^4 and reaches 7× at deg $= 3^{12} = 531441$ — the RF= 6/RP= 6 per-node shape — with the margin growing as n increases. It is therefore the enabler at the $R_F = 6$ gate; at the current $R_F = 4, R_P = 5$ frontier (deg $= 3^9 = 19683$) Euclidean is still faster, so that frontier stays on the schoolbook path. Lowering the crossover (base-case tuning, fewer allocations) is the remaining refinement.

- **Custom-MDS attack surface.** The organizer clarification makes MDS choice an admissible attack parameter, provided the matrix satisfies the invariant-subspace requirements. The current executable witnesses and economics are for the default Cauchy/Toeplitz instance only. A disciplined next experiment is therefore a reduced-round custom-MDS scout over Cauchy variants, circulant/Toeplitz families, and diagonal-plus-low-rank families, gated by the reference `verify_mds_matrix` check and measuring not only eliminant degree but also support growth, bivariate coefficient density, build memory, factor rate, and root-to-witness acceptance. A vulnerable admissible family that preserves degree but reduces the dense build by a large factor would be a genuine new attack surface; we do not claim such a family here.

- **Confidential frontier.** The density/zero-test records are confidential; a higher unrevealed submission cannot be excluded.

- **Official-verifier gate.** Witnesses should be re-checked through the official reference verifier at a pinned commit before any submission.

11 Conclusion

We have mapped the concrete-complexity landscape of the 2026 Poseidon1/KoalaBear challenge suite: a verifier-exact reimplementation for the default Cauchy/Toeplitz+Grain instance, a reproduced CICO-2 degree law on that verifier, a new root-first resultant model that puts the zero-test eliminant in CICO order with exact reduced-round witnesses (through $R_F = 4, R_P = 5$), a corrected density baseline with measured uniformity, and a full-strength collision no-go table. **We do not claim a solve of any full-round ($R_F = 6$) instance.** We measure the cost of reaching one and *port* the amortized resultant of [4] to this Poseidon1 zero-test route, validating it exactly at reduced rounds; priced end-to-end, however, the build-once bivariate construction dominates the per-node resultant and leaves the $R_F = 6$ route far outside practical reach under the current implementation model: on the order of 10^3 core-years and 1 TiB at $R_F = 6/R_P = 6$, growing to hundreds of TiB and $\sim 10^8$ core-years at the $R_P = 9$ target beyond the current public record. The load-bearing open question is therefore not merely faster resultants, but whether a custom admissible MDS, a blockwise build, or a different algebraic formulation can reduce the dense build itself.

References

[1] L. Grassi, D. Khovratovich, C. Rechberger, A. Roy, M. Schofnegger. *Poseidon: A New Hash Function for Zero-Knowledge Proof Systems.* ePrint 2019/458.

[2] L. Grassi, R. Lüftenegger, C. Rechberger, D. Rotaru, M. Schofnegger. *On a Generalization of Substitution-Permutation Networks: The HADES Design Strategy.* ePrint 2019/1107 (Eurocrypt 2020).

[3] Z. Zhao, J. Ding. *Breaking Poseidon Challenges with Graeffe Transforms and Complexity Analysis by FFT Lower Bounds.* ePrint 2025/950.

[4] A. Bak, A. Bariant, A. Bœuf, M. Hostettler, G. Jazeron. *Claiming bounties on small scale Poseidon and Poseidon2 instances using resultant-based algebraic attacks.* ePrint 2026/150.

[5] L. Grassi, C. Rechberger, M. Schofnegger, D. Khovratovich. *Survey of attacks on Poseidon and Poseidon2.* Version dated 22 November 2024, linked from the Poseidon Cryptanalysis Initiative.

[6] L. Grassi, K. Koschatko, C. Rechberger. *Poseidon and Neptune: Gröbner Basis Cryptanalysis Exploiting Subspace Trails.* ePrint 2025/954 (TOSC 2025).

[7] Z. Zhao, A. Sanso, G. Vitto, J. Ding. *Graeffe-Based Attacks on Poseidon and NTT Lower Bounds.* ePrint 2025/1916.

[8] Ethereum Foundation. *Poseidon Cryptanalysis Initiative.* `https://www.poseidon-initiative.info/`. Accessed 2026-06-17.